

Varsha Rajesh

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EDUCATION

University of Michigan, Ann Arbor, College of Engineering

Expected Graduation: Apr 2027

Bachelor of Science in Engineering, Data Science; Mathematics Minor

GPA: 3.92/4.00

- Relevant Coursework: Data Structures and Algorithms, Computational Methods in Statistics and Data Science, Applied Regression Analysis, Introduction to Statistical Computing, Discrete Mathematics, Database Management Systems

TECHNICAL SKILLS

Programming: Python, Pandas, NumPy, R, SQL, PySpark, MATLAB, C++, C, Assembly

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, CNNs, Grad-CAM, MLOps, NLP, Artificial Intelligence

Data & Analytics: Power BI, Tableau, A/B Testing, Hypothesis Testing, EDA, Data Cleaning, Insight Presentation, Database Management, Dashboard Creation, Trend Analysis, Predictive Analytics, ETL Processes, Statistical Techniques

Tools: MS365, Microsoft Office, Excel, Git, Azure DevOps, LLM/RAG Workflow, CVAT, Streamlit, Jupyter Notebooks, GDB

Methods: Feature Engineering, Classification and Regression Models, Data Transformation, Data Visualization, Statistical Modeling, Version Control, Problem Solving, Communication, Analytical Skills, Quality Assurance

PROFESSIONAL EXPERIENCE

Surgical Data De-Identification for AI Models Research Intern

July 2025 – Present

Michigan Medicine, Department of Cardiac Surgery and Robotics

Ann Arbor, MI

- Prepared and **de-identified sensitive clinical data** to enable privacy-preserving deployment of AI models in healthcare.
- Built **high-quality labeled datasets** supporting AI-driven analysis of cardiac surgery procedures in **ongoing clinical trials**.
- Improved annotation efficiency through close collaboration with researchers, accelerating safe and reliable AI model training.

Assessing Bioinformatics Capabilities of OpenAI GPT Research Intern

Aug 2024 – Apr 2025

University of Michigan Medical School, Department of Learning Health Sciences

Ann Arbor, MI

- Led a benchmark evaluation of **GPT-3.5, GPT-4o, and LLaMA-3-70B** across **104 bioinformatics tasks**, comparing outputs against **110–68,760 human solvers per task**.
- Built an end-to-end evaluation pipeline in Python, R, and PySpark to analyze model performance and systematic failures.
- Discovered strong alignment between LLM performance and human task difficulty while highlighting strengths and weaknesses of various LLM products; results under peer review and presented at a research symposium.

PROJECT EXPERIENCE

Alzheimer's MRI Classifier Project Team Member

Sept 2025 – Dec 2025

Michigan Data Science Team

Ann Arbor, MI

- Co-led a **7-member team** developing deep learning models in Python for early Alzheimer's detection from brain MRI scans, achieving **90.4% classification accuracy** with custom CNNs.
- Optimized VGG-16 and ResNet-50 architectures and designed ensemble strategies to improve predictive robustness.
- Applied Grad-CAM interpretability techniques to validate model decision-making in clinically relevant brain regions.
- Designed and deployed an end-to-end ML solution using Streamlit, operationalizing trained models for real-time MRI uploads, inference, and comparison against ground-truth labels, reflecting **MLOps** practices in deployment and iteration.

Stock Market Simulation

Sept 2025 – Oct 2025

Data Structures and Algorithms

Ann Arbor, MI

- Built a production-style C++ electronic exchange processing **tens of thousands of orders**, implementing price-time priority.
- Executed real-time analytics over trade data, including trader summaries and optimal time-traveler profit scenarios.
- Validated correctness and performance under adversarial workloads, meeting strict runtime and memory constraints.

Forum Test Classification: Machine Learning

Mar 2025 – Apr 2025

Applied NLP for Customer Signal Analysis

Ann Arbor, MI

- Implemented a Bernoulli Naive Bayes text classifier in C++ to categorize **5000+ posts** using a bag-of-words NLP model.
- Trained label priors and word-label likelihoods from historical data to identify recurring patterns and topic distributions.
- Designed log-probability scoring with smoothing to handle sparse terms, ensuring stable predictions in real-world data.
- Applied deterministic inference and performance optimization, enabling scalable analysis of large text datasets.
- Leveraged how structured text analytics can surface reliability signals from unstructured feedback, analogous to analyzing customer complaints for durability forecasting.

LEADERSHIP

Predicting Economic Values Project Lead

Dec 2025 – Present

Michigan Data Science Team

Ann Arbor, MI

- Led an end-to-end economics data science project for a **16-student team**, covering the full lifecycle from raw data extraction to predictive modeling and presentation.
- Developed a structured curriculum for team members spanning EDA, SQL, feature engineering, and regression modeling.
- Guided team members through Python-based regression, train-test splits, correlation analysis, and data storytelling.

Finance and Budgeting Lead

June 2025 – Present

Michigan Data Science Team

Ann Arbor, MI

- Managed, audited, and reconciled finances for a multi-project student organization, maintaining **over \$10,000**.
- Integrated **Excel-based financial models** into **Power BI** using Power Query for automated budget tracking.
- Built exploratory **Tableau** dashboards to visualize spending trends and category-level variance for leadership review.